

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

Senior engineer, 5+ years in React/TypeScript, Node/NestJS and AWS. I build and run production AI agents used by 50+ business customers, with the tests and monitoring behind them. Lead a team of 5. AI engineering since 2024.

Skills

Frontend: React 18/19, Next.js (SSR), TypeScript, Redux Toolkit, Module Federation, Nx, Vitest

Backend & Data: Python (FastAPI), Node.js, TypeScript, NestJS, PostgreSQL, Hasura (GraphQL), REST APIs

AI / GenAI: LLM Agents, Tool Calling, LangGraph, Multi-Agent (Supervisor, Swarm), RAG, Embeddings and Vector Search (OpenSearch kNN), Prompt Engineering, LiteLLM, LLM Evaluation, Guardrails, LLM Cost Tracking, Langfuse, OpenTelemetry, OCR / Vision

Cloud & DevOps: AWS (ECS, ECR, Lambda, S3, Amplify), Docker, GitHub Actions, CloudWatch

Engineering: System Design, Multi-Tenant SaaS, Evals, Feature Flags

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack)

May 2023 – Present

Hyderabad. B2B WhatsApp-commerce SaaS; customers include Sun Pharma, UltraTech, ITC, Johnson & Johnson, Veeba

- Led a team of 3 that replaced UltraTech's offline pour and dispatch forms with a WhatsApp workflow, **live in 2.5 weeks** (agent, backend, web app); **3,800+ pour cards and 1,900+ dispatches** captured across 200+ projects since May
- Redesigned the WhatsApp order agent (LangGraph) into **seven testable steps** on a visual builder, so customer-specific changes ship without code or a redeploy; running for Veeba, Noida Bakers and field ops
- Designed the multi-agent layer of the WhatsApp sales agent (LangGraph): messages route between service and payment agents, and a guardrail classifier filters spam, auto-replies and off-topic chat before the model answers
- Built automated quality tests for the order agent (**64 real scenarios**, run twice daily) that repeatedly caught breakages from other teams' backend changes, so fixes shipped before production broke
- Improved product matching on Veeba's **15,242 production aliases**: the correct first match rose from **69% to 78%** in testing and from **81% to 95%** on real rep messages; shipped behind an instant off switch
- Built end-to-end tracing (OpenTelemetry, Langfuse) so a failed AI order is debugged on one screen instead of in server logs
- Fixed the agent's document search so customers with many duplicate files get relevant answers, by searching only each customer's own documents (OpenSearch)
- Fixed a production crash loop in the messaging service and ended duplicate WhatsApp sends caused by repeated database updates, tracing both to root cause
- Own the order backend (NestJS, PostgreSQL; 50+ functions on one shared contract): cut product-search time **4x** (875 to 228 ms per line)
- Cut first page load from **15–30 seconds to about 1.5 seconds** across 19 apps (removing unused mobile code alone made builds **30–40% faster**), plus a custom bundler config and lazy loading
- Lead **5 engineers**: turn customer requirements into designs, review code, and own delivery and quality across 19 apps; write the design docs and incident reviews that let others pick up the work

Sumeru Software – Senior Developer

Aug 2022 – Jan 2023

- Built a role-based CRM from scratch (React, Redux Toolkit, Node.js) with permission-based screens and modular dashboards; owned it from kickoff to production and shipped the **MVP in 4 months** with a team of 4

Aerchain – Software Development Engineer

Intern Jul 2019 – Jul 2020; SDE 1 Apr 2021 – Jan 2022

- Built the real-time auction module for a B2B procurement platform, with an optimistic UI that reconciles with the server so bids feel instant and stay correct
- Built barcode-scan inventory management as an offline-first PWA (service workers) that keeps working without a connection

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020